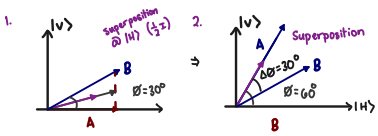
Homework 6

1. (2 pts) Unpolarized light is sent into a sequence of three polarizers, with the "easy" (transmission) axes at 0º, 30 º, and 60º, in that order.

$I_1 \quad I_2 \quad I_3$

a) If the initial intensity of the light is I, what is the intensity of the final output light?



$I_f = I\cos^2(\emptyset)$

$\rightarrow I_1 = I\cos^2(0°) = \frac{1}{2}I$

$\rightarrow I_2 = I_1\cos^2(30°) = \frac{3}{4}I_1 = \frac{3}{4}\left(\frac{1}{2}I\right)$

$\rightarrow I_3 = I_2\cos^2(30°) = \frac{3}{4}\left[\frac{3}{4}\left(\frac{1}{2}I\right)\right] \Rightarrow \left[\frac{3}{4}\cdot\frac{3}{4}\cdot\frac{2}{4}\right]I = \frac{18}{64}I = \frac{9}{32}I$

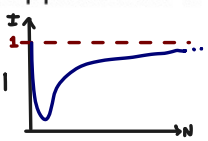
b) What is the polarization state of the output light—that is, what is its polarization direction?

$\underline{|60°\rangle} = \cos(60°|0°\rangle + \sin(60°|90°\rangle$

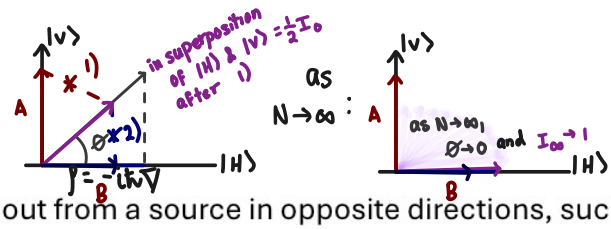
2. (3 pts) Show that if there are N polarizers one after the other, each with an angle of $\pi/2N$ relative to the one before it, that in the limit of $N \to \infty$, 100% of the light is transmitted. You can either do this by calculating a limit, or by plotting the result as a function of N and showing that it approaches 1.

$I = \lim_{N\to\infty}\left(\cos^2\left(\frac{\pi}{2N}\right)\right)^{N-1} =$

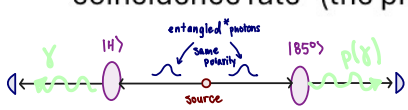
So, $I_\infty = 1$



3. (2 pts)



a) A correlated pair of photons is sent out from a source in opposite directions, such that the photons both have the same polarization, but the polarization direction can be any angle. If one of the photons passes through a polarizer at angle 0º ("H"), what is the probability that the other photon passes through a polarizer set at 85º? What is the "coincidence rate" (the probability that both pass through their polarizers) in this case?



a. $P(H|85) = \cos^2(H-85°) \Rightarrow \cos^2(85) \simeq .76\%$

b. $P(\text{both}) = \frac{1}{\pi}\left[\int_0^\pi \cos^2(\emptyset)d\emptyset \cdot \int_0^\pi \cos^2(\emptyset-85)d\emptyset\right]$

$*\cos^2(x) = \frac{1}{2}(1+\cos 2x)$: $\frac{1}{\pi}\left[\int_0^\pi \frac{1}{2}[1+\cos 2\emptyset]\cdot\frac{1}{2}[1+\cos(2\emptyset-170)]\,d\emptyset\right]$

$\hookrightarrow = \frac{1}{\pi}\int_0^\pi \frac{1}{4}\left(1+\cos(2\emptyset-170) + \cos 2\emptyset + \cos 2\emptyset(\cos(2\emptyset-170))\right)d\emptyset$

$\hookrightarrow \frac{1}{4\pi}\left[\int_0^\pi 1\,d\emptyset + \int_0^\pi \cos(2\emptyset-170)d\emptyset + \int_0^\pi \cos 2\emptyset\,d\emptyset + \int_0^\pi \cos 2\emptyset(\cos(2\emptyset-170))\,d\emptyset\right]$

$\hookrightarrow \pi \qquad \hookrightarrow 0 \qquad \hookrightarrow 0$

$\hookrightarrow \frac{1}{4\pi}\left[\int_0^\pi \cos 2\emptyset \cdot \cos(2\emptyset-170)\,d\emptyset\right] \underset{*}{\Rightarrow} \frac{1}{4\pi}\int_0^\pi \cos(2\emptyset-(2\emptyset-170)) + \cos(2\emptyset+(2\emptyset-170))d\emptyset$

$*\cos A \cdot \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$ $\hookrightarrow \frac{1}{4\pi}\left[\int_0^\pi \cos(170)\,d\emptyset + \int_0^\pi \cos(4\emptyset-170)d\emptyset\right]$ $\hookrightarrow 0$

$\hookrightarrow \frac{1}{4\pi}\int_0^\pi \cos(4\emptyset-170)\,d\emptyset \Rightarrow \frac{\pi}{2}\cos(170) \Rightarrow \frac{1}{4\pi}\left[\pi + \frac{\pi}{2}\cos(170)\right] = \frac{1}{4}\left[1+\frac{1}{2}\cos(170)\right] = 0.1268$ or $\simeq 12.68\%$

b) Suppose instead that the photons are sent out with either both exactly horizontal or both exactly vertical polarization, but randomly with 50% chance of either polarization. What is the probability that both photons pass through their polarizers in this case?

2 cases: $\emptyset = 0°$ ($|H\rangle$) - γ_1 through $|H\rangle \Rightarrow \cos^2(0°) = 1$

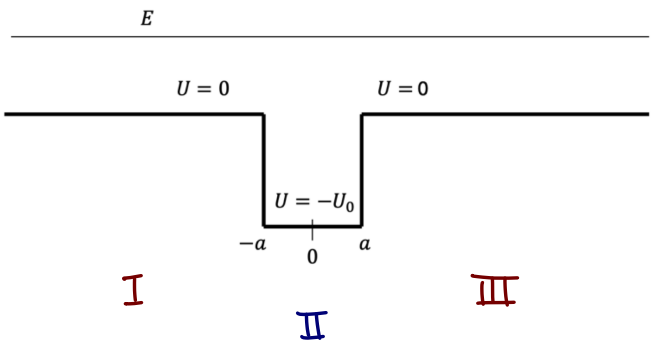
γ_2 through $|85\rangle \Rightarrow \cos^2(85°) = 0.0076$

Both $= 1 \cdot 0.0076 = 0.76\%$

$\emptyset = 90°$ - γ_1 through $|H\rangle = 0$

So $P_{final} = \frac{1}{2}(0 + 0.0076) = 0.0038 = 0.38\%$

4. (3 pts) In class we discussed non-bound states being scattered by an atom. Analyze the following case shown in the following picture, for an electron plane wave with kinetic energy E coming from the left and hitting a simple model of an atom with negative potential energy. As we have done in the past, write down a set of equations you would solve to get the transmission and reflection coefficients, but don't solve them.



$$\mathcal{E}\psi(x) = -\frac{\hbar}{2m}\frac{d^2}{dx}\psi(x) + U(x)\psi(x)$$

$$Ae^{-kx} + Be^{kx}$$

Reflection Coefficient $R = |r|^2$

outside the well: $k = \frac{\sqrt{2mE}}{\hbar}$

inside the well: $q = \frac{\sqrt{2m(E+U_0)}}{\hbar}$

$$U(x)\begin{cases} \text{I}: U:0,\ x<-a & - \ \psi_1 = Ae^{ikx} + Be^{-ikx} \\ \text{II}: U:-U_0,\ -a<x<a & - \ \psi_2 = Ce^{iqx} + De^{-iqx} \\ \text{III}: U:0,\ a<x & - \ \psi_3 = Fe^{ikx} \end{cases}$$

@ $-a$: $\psi_1 = \psi_2$: $Ae^{-ika} + Be^{ika} = Ce^{-iqa} + De^{iqa}$

and $\frac{d\psi_1}{dx} = \frac{d\psi_2}{dx}$: $ikAe^{-ika} - ikBe^{ika} = iqCe^{-iqa} - iqDe^{-iqa}$

@ a: $\psi_2 = \psi_3$: $Ce^{iqx} + De^{-iqx} = Fe^{ikx}$

and $\frac{d\psi_2}{dx} = \frac{d\psi_3}{dx}$: $iqCe^{iqx} - De^{-iqx} = ikFe^{ikx}$

$$R = \left|\frac{B}{A}\right|^2, \quad T = \left|\frac{F}{A}\right|^2$$